Year 11 · AQA Level 2 Further Maths · Covered topic

Surds: simplify, expand and rationalise

Revisit and practise · OnMaths · Created by Mr Ali

Today I will learn to...

Goals

1. Simplify and calculate with surds

2. Rationalise the denominator

Do now: retrieval practice

4 mins

1 Work out $\sqrt{49} \times \sqrt{4}$

14

2 Write 72 as a product of prime factors

$2^3 \times 3^2$

3 Expand $(x + 3)(x - 5)$

$x^2 - 2x - 15$

4 Expand $(a + b)(a - b)$

$a^2 - b^2$

Objective 1

Simplify and calculate with surds

Key idea

Learn

!

$\sqrt{ab} = \sqrt{a} \times \sqrt{b}$. To simplify, find the largest square factor. Like surds collect just like algebra: $3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$. Also $\sqrt{a} \times \sqrt{a} = a$.

Objective 1: Simplify and calculate with surds

I do: example 1

I do

Simplify $\sqrt{72}$

1. Find the largest square factor of 72: it is 36
2. $\sqrt{72} = \sqrt{36} \times \sqrt{2}$
3. $= 6\sqrt{2}$

I do: example 2

I do

Simplify $\sqrt{50} + \sqrt{18}$

1. $\sqrt{50} = \sqrt{25} \times \sqrt{2} = 5\sqrt{2}$
2. $\sqrt{18} = \sqrt{9} \times \sqrt{2} = 3\sqrt{2}$
3. $5\sqrt{2} + 3\sqrt{2} = 8\sqrt{2}$

I do: example 3

I do

Expand and simplify $(3 + \sqrt{5})(2 - \sqrt{5})$

1. $3 \times 2 = 6$
2. $3 \times (-\sqrt{5}) = -3\sqrt{5}$ and $\sqrt{5} \times 2 = 2\sqrt{5}$
3. $\sqrt{5} \times (-\sqrt{5}) = -5$
4. $6 - 3\sqrt{5} + 2\sqrt{5} - 5 = 1 - \sqrt{5}$

Mini whiteboards

We do

1. Simplify $\sqrt{48}$

Answer: $4\sqrt{3}$

2. Expand and simplify $(\sqrt{3} + 1)^2$

Answer: $4 + 2\sqrt{3}$

3. Simplify $\sqrt{75}$

Answer: $5\sqrt{3}$

4. Simplify $2\sqrt{12} + \sqrt{27}$

Answer: $4\sqrt{3} + 3\sqrt{3} = 7\sqrt{3}$

You do: choose your challenge

You do

Bronze

Simplify $\sqrt{20}$

Silver

Simplify $3\sqrt{8} + \sqrt{32}$

Gold

Expand and simplify $(4 - \sqrt{3})(4 + \sqrt{3})$

Platinum

A rectangle measures $(3 + \sqrt{2})$ cm by $(3 - \sqrt{2})$ cm. Find its area and the length of its diagonal.

You do: answers

Check

Bronze

$2\sqrt{5}$

Silver

$6\sqrt{2} + 4\sqrt{2} = 10\sqrt{2}$

Gold

$16 - 3 = 13$

Platinum

Area = $9 - 2 = 7$ cm^2. Diagonal^2 = $(11 + 6\sqrt{2}) + (11 - 6\sqrt{2}) = 22$, so the diagonal is $\sqrt{22}$ cm

If you finish early

Extra

Bronze

Simplify $\sqrt{32}$

$4\sqrt{2}$

Silver

Simplify $\sqrt{18} + \sqrt{50}$

$3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$

Gold

Expand and simplify $(2 + \sqrt{5})(3 - \sqrt{5})$

$6 - 2\sqrt{5} + 3\sqrt{5} - 5 = 1 + \sqrt{5}$

Platinum

Show that $(\sqrt{12} + \sqrt{3})^2 = 27$

$\sqrt{12} = 2\sqrt{3}$, so $(3\sqrt{3})^2 = 27$

Exam question 1

2 marks

Show that $\sqrt{75} - \sqrt{12} = 3\sqrt{3}$

Space for working

Mark scheme: $\sqrt{75} = 5\sqrt{3}$ and $\sqrt{12} = 2\sqrt{3}$ (M1). $5\sqrt{3} - 2\sqrt{3} = 3\sqrt{3}$ (A1)

Exam question 2

3 marks

Expand and simplify $(2\sqrt{3} + 1)(\sqrt{3} - 4)$

Space for working

Mark scheme: $6 - 8\sqrt{3} + \sqrt{3} - 4$ (M1 four terms, M1 $2\sqrt{3} \times \sqrt{3} = 6$) $= 2 - 7\sqrt{3}$ (A1)

Objective 1 achieved

Simplify and calculate with surds



Objective 2

Rationalise the denominator

✓ Achieved: Simplify and calculate with surds

Key idea

Learn

!

Rationalising removes the surd from the bottom of a fraction. For $a/\sqrt{b}$, multiply top and bottom by $\sqrt{b}$. For $a/(b + \sqrt{c})$, multiply top and bottom by $(b - \sqrt{c})$: the difference of two squares clears the surd.

Objective 2: Rationalise the denominator

I do: example 1

I do

Rationalise $6/\sqrt{3}$

1. Multiply top and bottom by $\sqrt{3}$
2. $6\sqrt{3} / 3$
3. $= 2\sqrt{3}$

I do: example 2

I do

Rationalise $10/\sqrt{5}$

1. Multiply top and bottom by $\sqrt{5}$
2. $10\sqrt{5} / 5$
3. $= 2\sqrt{5}$

I do: example 3

I do

Rationalise $4/(3 - \sqrt{5})$

1. Multiply top and bottom by $(3 + \sqrt{5})$
2. Bottom: $(3 - \sqrt{5})(3 + \sqrt{5}) = 9 - 5 = 4$
3. Top: $4(3 + \sqrt{5})$
4. $4(3 + \sqrt{5}) \div 4 = 3 + \sqrt{5}$

Mini whiteboards

We do

1. Rationalise $12/\sqrt{6}$

Answer: $2\sqrt{6}$

2. Rationalise $1/(\sqrt{2}+1)$

Answer: $\sqrt{2}-1$

3. Rationalise $6/\sqrt{2}$

Answer: $3\sqrt{2}$

4. Rationalise $4/(\sqrt{3}-1)$

Answer: $2\sqrt{3}+2$

You do: choose your challenge

You do

Bronze

Rationalise $9/\sqrt{3}$

Silver

Rationalise $5/(2\sqrt{5})$

Gold

Rationalise $2/(\sqrt{5} - 1)$

Platinum

Write $(1 + \sqrt{2})/(3 - \sqrt{2})$ in the form $a + b\sqrt{2}$

You do: answers

Check

Bronze

$3\sqrt{3}$

Silver

$5\sqrt{5}/10 = \sqrt{5}/2$

Gold

$2(\sqrt{5} + 1)/4 = (\sqrt{5} + 1)/2$

Platinum

$(1 + \sqrt{2})(3 + \sqrt{2})/7 = (5 + 4\sqrt{2})/7 = 5/7 + 4/7\ \sqrt{2}$

If you finish early

Extra

Bronze

Rationalise $8/\sqrt{2}$

$4\sqrt{2}$

Silver

Rationalise $3/(2\sqrt{3})$

$\sqrt{3}/2$

Gold

Rationalise $6/(\sqrt{7} + 1)$

$6(\sqrt{7} - 1)/6 = \sqrt{7} - 1$

Platinum

Write $(3 + \sqrt{5})/(3 - \sqrt{5})$ in the form $a + b\sqrt{5}$

$(14 + 6\sqrt{5})/4 = 7/2 + 3/2\sqrt{5}$

Exam question 1

2 marks

Write $14/\sqrt{7}$ in the form $a\sqrt{7}$

Space for working

Mark scheme: $14\sqrt{7} / 7$ (M1) = $2\sqrt{7}$ (A1)

Exam question 2

3 marks

Write $(5 + \sqrt{3})/(2 - \sqrt{3})$ in the form $a + b\sqrt{3}$

Space for working

Mark scheme: Multiply by $(2 + \sqrt{3})/(2 + \sqrt{3})$ (M1). Top: $10 + 5\sqrt{3} + 2\sqrt{3} + 3 = 13 + 7\sqrt{3}$, bottom: $4 - 3 = 1$ (M1). $13 + 7\sqrt{3}$ (A1)

Objective 2 achieved

Rationalise the denominator



Real life

Apply

A square TV screen has a diagonal of 40 cm. Find its side length as a simplified surd.

$s\sqrt{2} = 40$, so $s = 40/\sqrt{2} = 20\sqrt{2}$ cm

Extension: stretch yourself

Stretch

1. Show that $(\sqrt{8} + \sqrt{18})^2 = 50$

$\sqrt{8} = 2\sqrt{2}$ and $\sqrt{18} = 3\sqrt{2}$, so the sum is $5\sqrt{2}$. $(5\sqrt{2})^2 = 25 \times 2 = 50$

2. Solve $x\sqrt{3} = x + 2$. Give x in the form $a + b\sqrt{3}$

$x(\sqrt{3} - 1) = 2$, $x = 2/(\sqrt{3} - 1) = 2(\sqrt{3} + 1)/2 = 1 + \sqrt{3}$

Exit ticket: exam questions

Exit

1. Simplify $\sqrt{98}$ **(1 mark)**

2. Expand and simplify $(3 + \sqrt{2})^2$ **(2 marks)**

3. Write $8/(3 + \sqrt{5})$ in the form $a + b\sqrt{5}$ **(2 marks)**

Exit ticket: mark scheme

Mark it

Q1 $7\sqrt{2}$ [1]

Q2 $9 + 6\sqrt{2} + 2 = 11 + 6\sqrt{2}$ [2]

Q3 $8(3 - \sqrt{5})/(9 - 5)$ (M1) $= 6 - 2\sqrt{5}$ (A1) [2]

Total: ____ / 5

Today I achieved

- ✔ **Simplify and calculate with surds**
- ✔ **Rationalise the denominator**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Simplify $\sqrt{200}$ [1]

2. Simplify $\sqrt{45} + \sqrt{80}$ [2]

3. Expand and simplify $(2 + \sqrt{7})(3 - \sqrt{7})$ [2]

4. Rationalise $15/\sqrt{5}$ [2]

5. Write $(4 + \sqrt{2})/(4 - \sqrt{2})$ in the form $a + b\sqrt{2}$ [3]

Extra practice: Mr Barton Maths: AQA Further Maths → search "Surds"